02 | Performance changes with the data source

All registered 20% and 100% dense-stage cells; each number is the mean of seeds 11, 22 and 33

20% budget | Image AUROC

	Internal	ARTS	CFVS	CSIR
Adapted CNN	0.814	0.569	0.624	0.829
DINOv3 frozen	0.790	0.592	0.847	0.608
DINOv3 LoRA	0.936	0.825	0.982	0.891
DINOv3 full	0.927	NA	NA	NA
DINOv3 + reference	0.783	0.619	0.694	0.587
Reference: no position	0.804	NA	NA	NA
Reference: mean evidence	0.768	NA	NA	NA

20% budget | Native pixel AP

	Internal	ARTS	CFVS	CSIR
Adapted CNN	0.224	0.042	0.180	0.207
DINOv3 frozen	0.167	0.035	0.120	0.082
DINOv3 LoRA	0.200	0.053	0.149	0.082
DINOv3 full	0.214	NA	NA	NA
DINOv3 + reference	0.206	0.041	0.101	0.077
Reference: no position	0.199	NA	NA	NA
Reference: mean evidence	0.189	NA	NA	NA

20% budget | Crack-positive Dice

	Internal	ARTS	CFVS	CSIR
Adapted CNN	0.155	0.103	0.106	0.190
DINOv3 frozen	0.094	0.055	0.139	0.143
DINOv3 LoRA	0.160	0.089	0.183	0.123
DINOv3 full	0.151	NA	NA	NA
DINOv3 + reference	0.138	0.070	0.061	0.140
Reference: no position	0.157	NA	NA	NA
Reference: mean evidence	0.102	NA	NA	NA

100% budget | Image AUROC

	Internal	ARTS	CFVS	CSIR
Adapted CNN	0.849	0.584	0.947	0.487
DINOv3 frozen	0.872	0.732	0.894	0.663
DINOv3 LoRA	0.961	0.882	0.991	0.922
DINOv3 full	0.961	NA	NA	NA
DINOv3 + reference	0.874	0.735	0.903	0.648
Reference: no position	0.879	NA	NA	NA
Reference: mean evidence	0.880	NA	NA	NA

100% budget | Native pixel AP

	Internal	ARTS	CFVS	CSIR
Adapted CNN	0.249	0.074	0.210	0.101
DINOv3 frozen	0.206	0.065	0.083	0.079
DINOv3 LoRA	0.224	0.072	0.135	0.071
DINOv3 full	0.229	NA	NA	NA
DINOv3 + reference	0.208	0.069	0.068	0.067
Reference: no position	0.199	NA	NA	NA
Reference: mean evidence	0.230	NA	NA	NA

100% budget | Crack-positive Dice

	Internal	ARTS	CFVS	CSIR
Adapted CNN	0.198	0.133	0.238	0.077
DINOv3 frozen	0.112	0.082	0.059	0.154
DINOv3 LoRA	0.149	0.093	0.175	0.142
DINOv3 full	0.147	NA	NA	NA
DINOv3 + reference	0.139	0.097	0.050	0.141
Reference: no position	0.144	NA	NA	NA
Reference: mean evidence	0.171	NA	NA	NA

NA: this system was not registered for that condition. Missing cells are not zero and are not imputed.

All panels use the same 0-1 scale. AP depends on cohort composition and annotation; colors are not a cross-cohort ranking.

ARTS, CFVS and CSIR are observed acquisition-held-out development populations. These panels do not show final confirmation outcomes.